Multimedia Signal Processing - 2nd Module

February, 19 2018

1. **Question (8 pts)**
   Consider a continuous source X that emits symbols following a probability distribution function (p.d.f.) $p_X(x)$.
   (a) Consider the following series of symbols emitted by X

   $$x = \{-0.8,\ 0.1,\ -0.8,\ -0.7,\ 0.7,\ -0.7,\ -0.2,\ 0.1,\ 0.8,\ -0.2,\ 1.2\}.$$

   Design a Lloyd-Max quantizer tailored to these symbols using $M = 3$ bins and the thresholds $\{-\infty, -0.5, 0.5, +\infty\}$ as initialization. Stop after one complete iteration.
   (b) How many iterations must be performed to reach the algorithm's convergence?
   (c) Quantize x using the Lloyd-Max quantizer designed at point (b), thus obtaining $\tilde{x}$.
   (d) Compute the entropy of the source representing $\tilde{x}$ estimating the probability mass function (p.m.f.) directly from $\tilde{x}$.
   (e) Derive an Huffman code to transmit $\tilde{x}$.
   (f) Represent using Huffman code a string of three symbols of your choice.

2. **Question (5 pts)**
   (a) Draw the JPEG image encoder block-diagram.
   (b) Report a step-by-step explanation of how to encode and decode an image using JPEG.

3. **Question (5 pts)**
   (a) Describe the concepts of absolute threshold of hearing and acoustic masking, explaining why they are important in the audio coding field.
   (b) Report the steps to compute a perceptual audio model using MPEG-1 psychoacoustic model 1.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a typical hybrid DPCM/DCT video coding scheme.
   - [T] [F] Prediction methods are used only for inter-coded frames.
   - [T] [F] It is possible to compute motion vectors with precision smaller than one pixel.
   - [T] [F] Video frames can be decoded in any random order decided by the decoder.
   - [T] [F] When increasing the block size, the energy of the prediction residuals increases.

   (b) Consider the problem of speech coding.

- [T] [F] The spectral envelop of unvoiced signals is characterized by a set of formants.
- [T] [F] When using a DPCM, the long-term predictor helps compensating the pitch structure of the voiced spectrum.
- [T] [F] When using linear predictive vocoder exploiting the source-system model, the coefficients of the synthesis filter are directly quantized.

(c) Consider two discrete memoryless sources X and Y emitting symbols from a finite alphabet.

- [T] [F] $H(X)$ is the minimum amount of bits that can be used to encode symbols of X without introducing distortion.
- [T] [F] It is not possible to encode symbols of X with a rate greater than $H(X) + 1$.
- [T] [F] $H(X) + H(Y) = H(X;Y)$ if the two sources are independent.

5. **Question (Matlab - 10pts)**

You are given a music audio recording `sample.wav` that needs to be compressed in a lossy fashion with at most $R = 6$ bit/symbol. You decide to test several coding techniques in order to maximize the Signal-to-Noise Ratio that you can get with the given bitrate budget.

(a) Load the audio file and plot the corresponding signal.

(b) Determine the SNR using a linear PCM encoding scheme.

(c) Determine the SNR using an Open-Loop DPCM encoding scheme. Assume that the optimal predictor for each sample is the previous sample.

(d) Determine the SNR using an Closed-Loop DPCM encoding scheme. Assume that the optimal predictor for each sample is the previous sample.

(e) Draw on the same plot the reconstruction error over time between the original signal and

1. The PCM encoded signal.
2. The Opel-Loop DPCM signal reconstructed at the receiver.
3. The Close-Loop DPCM signal reconstructed at the receiver.

What is the relationship between the obtained SNR values and the plotted errors?

You decide that it is worth trying a transform coding scheme based on DCT transform, considering a block size $B = 64$ samples.

(f) Reshape the input signal as a 64×200 matrix.

(g) Transform each column of the matrix into its corresponding DCT representation.

(h) Quantize the transformation matrix with a PCM scheme using Optimal Bit Allocation (hint: $R_k = R + \frac{1}{2}\log_2\left(\frac{\sigma_{y_k}^2}{\left(\prod_{i=1}^{N}\sigma_{y_i}^2\right)^{1/N}}\right)$). First compute the number of bits for each of the 64 coefficients, then quantize each row of the matrix with the proper coefficient.

(i) Reconstruct the signal in the time domain and compute the SNR.

(j) Plot the allocated bits for each coefficient and comment the obtained result in terms of SNR.